



AIMS

African Institute for
Mathematical Sciences
SENEGAL

Project Report

Real LLM Deployment and Standard Industrial Methodologies

Wolof Educational Assistant — LoRA Fine-tuned Qwen3-0.6B

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GitHub:	https://github.com/mairamesniang12/ aims-gaai-exam-wolof-assistant
HF Model:	https://huggingface.co/niangmariame513/ wolof-assistant-qwen3
HF Space:	https://huggingface.co/spaces/niangmariame513/ wolof-assistant-demo
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1. Problem Definition

- **Use case:** Wolof educational assistant that answers basic questions and assists with language learning tasks in Wolof, the most widely spoken language in Senegal (over 10 million speakers).
- **Users:** Senegalese learners and speakers who want a digital assistant in their native language; researchers and students interested in low-resource African language NLP.
- **Why fine-tuning is needed:** The base model (Qwen3-0.6B) has very limited Wolof capability. Without fine-tuning, the model ignores Wolof input or responds in English. Fine-tuning on Wolof instruction-response pairs teaches the model to follow instructions in Wolof and produce relevant answers.
- **Why a small model is appropriate:** Qwen3-0.6B fits in memory on a single T4 GPU (15 GB VRAM) with LoRA, making it trainable within the compute constraints of Google Colab free tier. Only 1.67% of parameters are trainable (10,092,544 out of 606,142,464), enabling fast iteration with limited resources.

2. Data Preparation

We used three separated data sources. Each source was kept in a distinct file.

Table 1: Data sources used for training

aimsred				
CohereLabs/aya_dataset	500	Instruction following	Filtered to Wolof; max 500	General base
aimsgray	500	Text understanding	Kept as-is; orthography documented	Domain Wolof
soynade-research/Wolof-Non-Standard-Orthography				
synthetic_wolof_instructions	5300	Instruction following	No filtering needed	Synthetic

Every example was converted to **system/user/assistant** chat format:

Listing 1: Chat format conversion

```
{
  "role": "system",
  "content": "You are a helpful Wolof language assistant."
}
{
  "role": "user",
  "content": "<instruction> <input>"
}
{
  "role": "assistant",
  "content": "<output>"
}
```

Tokenization confirmation

avg_total=129.0, avg_supervised=33.9, kept=5670, skipped=0

3. Splitting Strategy

Table 2: Train/validation/eval split sizes

aimsred				
AYA	450	25	25	500
aimsgray	450	25	25	500
Soynade				
Synthetic	4770	265	265	5300
aimsgray Total	5670	315	315	6300

The split was performed **per source family** with a deterministic seed before merging, preventing data leakage: no example appears in both train and eval. The combined `eval_all.jsonl` (315 examples) was used for all automatic metrics.

4. Training Methodology

Base model choice: Qwen/Qwen3-0.6B fits on T4 GPU with LoRA, supports system/user/assistant chat templates natively, has multilingual capability, and is freely available on Hugging Face Hub.

Table 3: LoRA and training configuration

aimsred	
LoRA rank (r)	16
aimsgray LoRA alpha	32
LoRA dropout	0.05
aimsgray Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Trainable params	10,092,544 (1.67%)
aimsgray Epochs	3
Hardware	Google Colab T4 GPU (≈ 2.5 hours)

Assistant-only loss and -100 masking: In PyTorch `CrossEntropyLoss`, label -100 is the `ignore_index`: token contributes zero to loss, no gradient computed.

Listing 2: Label masking

```
system tokens    -> label = -100    # ignored
user tokens     -> label = -100    # ignored
padding tokens  -> label = -100    # ignored
assistant tokens -> normal token ids # trained on
```

Only **33.9/129.0 tokens (26%)** per example contributed to the loss.

Checkpoint policy: `best_adapter/` (min val loss) and `latest_adapter/` (end of training). Best adapter used for Hub upload.

Monitoring: Loss logged every 100 steps. Validation every ≈ 0.14 epochs. TensorBoard logs at `outputs/tensorboard/qwen3_0_6b_wolof_lora/`.

5. Evaluation

Listing 3: Real evaluation output

```
=== Aggregate metrics ===
exact_match: 0.3492
token_f1: 0.7499
bleu: 0.6665
rouge_l: 0.7456
```

Table 4: Automatic evaluation metrics on 315 held-out examples

aimsred		
Exact Match	0.3492 (34.9%)	34.9% match reference exactly
aimsgray Token F1	0.7499 (75.0%)	Strong token-level overlap
BLEU	0.6665 (66.7%)	Good n-gram precision for 6300 examples
aimsgray ROUGE-L	0.7456 (74.6%)	Good LCS recall

Table 5: Training and validation loss progression

aimsred		
0.14	5.536 → 1.886	1.556
aimsgray 0.56	≈0.35	0.261
1.00	≈0.21	0.196
aimsgray 1.55	≈0.18	0.162
2.00	≈0.09	0.145
aimsgray 2.26 (final)	≈0.037	0.148

✓ Partial success — geography

Q: Alger moy peeyu ban rew? Ref: Alger moy peeyu rewum Algerie

Pred: Alger moy peyum rewum Algiers F1=0.60, BLEU=0.39

Content correct; surface spelling errors.

✗ Failure — Wolof to French translation

Q: wagnil ma: benn, naar, nett... Ref: un, deux, trois...

Pred: Benn: Naa ngi ci, men na nekk... F1=0.00, BLEU=0.04

Model responded in Wolof instead of French.

✗ Failure — geography hallucination

Q: Yamoussoukro mooy geblag man rew? Ref: ...rewum Cote d'Ivoire.

Pred: Yamoussuksu ker gi "Rew?" ak "geblag."... F1=0.00, BLEU=0.09

Model hallucinated; did not answer the question.

6. Deployment

- **Hugging Face model repo:** <https://huggingface.co/niangmariame513/wolof-assistant-qwen3>
Files: `adapter_config.json`, `adapter_model.safetensors`, `tokenizer`. Visibility: Public.
- **Hugging Face Space:** <https://huggingface.co/spaces/niangmariame513/wolof-assistant-demo>
Status: **Running** (green). Loads from Hub, not local checkpoint.
- **Model card:** <https://huggingface.co/niangmariame513/wolof-assistant-qwen3>
Includes: 3 sources, LoRA config, real metrics (BLEU=0.6665, ROUGE-L=0.7456), loading example, limitations.
- **Required hardware:** CPU sufficient; GPU recommended for speed.

7. Limitations and Risks

- **Hallucination:** Plausible-sounding but wrong answers for OOD topics (Yamous-soukro: F1=0.00).
- **Poor categories:** Translation tasks (F1=0.00, BLEU=0.04) and some geography fail due to underrepresentation in training data.
- **Data bias:** Synthetic dominates (4770 vs 450+450). Soynade non-standard orthography causes spelling inconsistencies.
- **Unsafe prompts:** Keyword-based safety filter added but not robust. Model not evaluated for harmful Wolof outputs.
- **Prompt injection:** `context_state_machine.py` retrieval can be exploited. Mitigated by confidence score.
- **Whisper ASR:** Wolof not natively supported, limiting voice-based interface development.

8. What We Improved

- **Data quality:** Each source documented separately. Verified 5670/5670 examples passed tokenization (skipped=0). Soynade orthography documented as formal limitation.
- **src/context_state_machine.py:** Added (1) confidence score mechanism returning fallback when retrieval quality is low; (2) keyword-based safety filter for out-of-scope requests.
- **Deployment app:** Replaced `gr.Interface` and placeholder `MODEL_REPO_ID` with `gr.Blocks` loading from real Hub model, with Wolof prompts and visible guardrail section.

- **Evaluation:** Real automatic metrics + qualitative examples with actual scores, failure cases and root-cause analysis.

9. Individual Technical Notes

Student 1: Mairame Niang

- **Contribution:** Model training (Colab T4); HF Hub deployment; Gradio Space (`gr.Interface` → `gr.Blocks`); GitHub management.
- **Technical choice:** Colab T4 GPU over local CPU (41 min/100 examples on CPU vs 2.5h full training on T4). `gr.Blocks` over `gr.Interface` for structured layout.
- **Problem diagnosed:** `torchao v0.10.0` conflict; fixed with `pip install torchao==0.16.0`. HF token 401 error; fixed with new Write token and `hf auth login -force`.
- **Verification:** Loss 5.536→0.037 over 3 epochs; val loss 0.145; Space status Running; check-auth confirmed `niangmariame513`.
- **Next improvement:** Side-by-side comparison in Space: base Qwen3-0.6B vs fine-tuned on same Wolof prompt.

Student 2: Fatima Sane

- **Contribution:** Data preparation (`download_datasets.py`); 3 sources downloaded and separated; chat format verified; `data/README.md` written.
- **Technical choice:** Kept 3 separate files over merging into `wolof_all.jsonl`. Merging violates exam methodology and prevents per-source traceability.
- **Problem diagnosed:** `prepare` subcommand gave WARNING for missing `aya/soynade` files. Fixed by running `all: download_datasets.py all -max-aya-examples 500 -max-soynade-examples 500`.
- **Verification:** Script output: 500 aya rows, 500 soynade rows, 5300 synth rows, 315 eval rows. Training: `kept=5670`, `skipped=0`.
- **Next improvement:** Filter outputs <5 tokens, duplicates across sources, or non-Wolof text.

Student 3: Fatou Bintou

- **Contribution:** Evaluation on 315 examples (`evaluation.py`); real metrics (EM=0.35, F1=0.75, BLEU=0.67, ROUGE-L=0.75); model card written and uploaded to HF Hub.
- **Technical choice:** Reported BLEU + ROUGE-L over BLEU alone. ROUGE-L captures LCS recall, more informative for short Wolof outputs.
- **Problem diagnosed:** Systematic failures on translation (F1=0.00) and OOD geography (F1=0.00). Diagnosed as data imbalance (synthetic dominates; these task types underrepresented).
- **Verification:** Real output: `exact_match=0.3492`, `token_f1=0.7499`, `bleu=0.6665`,

rouge_l=0.7456. Model card live on HF.

- **Next improvement:** Per-source metrics (AYA vs Soynade vs Synthetic) to identify which source needs more data.

Student 4: Christian

- **Contribution:** Improved `context_state_machine.py` (confidence score + safety filter); qualitative testing of Space with 10 prompts.
- **Technical choice:** Keyword-based filter over classifier (avoids extra model, latency). Length-based confidence over cosine similarity (avoids CPU embedding overhead).
- **Problem diagnosed:** OOD prompts give confident wrong answers (F1=0.00 confirmed by evaluation). Model has no OOD detection.
- **Verification:** 5 safety prompts all returned guardrail. Cross- referenced: Yamoussoukro F1=0.00 (OOD), Alger F1=0.60 (in-distribution).
- **Next improvement:** Embedding-based OOD detector with `paraphrase-multilingual-MiniLM-L12` cosine threshold 0.4.

Student 5: Jean Baptiste

- **Contribution:** Project report (`PROJECT_REPORT_TEMPLATE.md`); 5-prompt demo script; submission checklist verification.
- **Technical choice:** 7-step pipeline structure over free-form essay (maps to rubric). Full loss curve table to demonstrate convergence.
- **Problem diagnosed:** First draft used estimated metrics. Corrected when real `evaluation.py` output (BLEU=0.6665, ROUGE-L=0.7456) became available.
- **Verification:** All 8 checklist items verified. Metrics confirmed: EM=0.3492, F1=0.7499, BLEU=0.6665, ROUGE-L=0.7456. 5 demo prompts tested on live Space.
- **Next improvement:** Add qualitative comparison (base model vs fine-tuned) to support the 15% training quality criterion.